

Solution Add row 1 to row 2. Then add row 2 to row 3. Then add row 3 to row 4:

First differences 1, -1
$$U = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now add row 3 to row 2. Then add row 2 to row 1:

Reduced form
$$R = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}.$$

The rank is $r = 3$. There is one free variable ($n - r = 1$). The special solution is $s = (1, 1, 1, 1)$. Every row adds to 0. Notice $-F = (1, 1, 1)$ in the pivot variables of s .

3.3 B Factor these rank one matrices into $A = uv^T = \text{column times row}$:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (\text{find } d \text{ from } a, b, c \text{ if } a \neq 0)$$

Split this rank two matrix into $u_1 v_1^T + u_2 v_2^T = (3 \text{ by } 2) \text{ times } (2 \text{ by } 4) \text{ using } R$:

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 \\ 2 & 3 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = E^{-1}R.$$

Solution For the 3 by 3 matrix A , all rows are multiples of $v^T = [1 \ 2 \ 3]$. All columns are multiples of the column $u = (1, 2, 3)$. This symmetric matrix has $u = v$ and A is uu^T . Every rank one symmetric matrix will have this form or else $-uu^T$.

If the 2 by 2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has rank one, it must be singular. In Chapter 5, its determinant is $ad - bc = 0$. In this chapter, row 2 is c/a times row 1.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 \\ c/a \end{bmatrix} \begin{bmatrix} a & b \end{bmatrix} = \begin{bmatrix} a & b \\ c & bc/a \end{bmatrix}. \quad \text{So } d = \frac{bc}{a}.$$

The 3 by 4 matrix of rank two is a sum of *two matrices of rank one*. All columns of A are combinations of the pivot columns 1 and 2. All rows are combinations of the nonzero rows of R . The pivot columns are u_1 and u_2 and those rows are v_1^T and v_2^T . Then A is $u_1 v_1^T + u_2 v_2^T$, multiplying r columns of E^{-1} times r rows of R :

Columns times rows
$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 \\ 2 & 3 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}$$

3.3 C Find the row reduced form R and the rank r of A and B (those depend on c). Which are the pivot columns of A ? What are the special solutions and the matrix N ?

Find special solutions $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 4 & 8 & c \end{bmatrix}$ and $B = \begin{bmatrix} c & c \\ c & c \end{bmatrix}$.

Solution The matrix A has rank $r = 2$ *except if* $c = 4$. The pivots are in columns 1 and 3. The second variable x_2 is free. Notice the form of R :

$$c \neq 4 \quad R = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad c = 4 \quad R = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Two pivots leave one free variable x_2 . But when $c = 4$, the only pivot is in column 1 (rank one). The second and third variables are free, producing two special solutions:

$$c \neq 4 \quad \text{Special solution with } x_2 = 1 \text{ goes into } N = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}.$$

$$c = 4 \quad \text{Another special solution goes into } N = \begin{bmatrix} -2 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The 2 by 2 matrix $\begin{bmatrix} c & c \\ c & c \end{bmatrix}$ has rank $r = 1$ *except if* $c = 0$, when the rank is zero!

$$c \neq 0 \quad R = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad N = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \text{Nullspace} = \text{line}$$

The matrix has *no pivot columns* if $c = 0$. Then both variables are free:

$$c = 0 \quad R = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{Nullspace} = \mathbf{R}^2.$$

Problem Set 3.3

1 Which of these rules gives a correct definition of the *rank* of A ?

- (a) The number of nonzero rows in R .
- (b) The number of columns minus the total number of rows.
- (c) The number of columns minus the number of free columns.
- (d) The number of 1's in the matrix R .

- 2 Find the reduced row echelon forms R and the rank of these matrices:

- (a) The 3 by 4 matrix with all entries equal to 4.
 (b) The 3 by 4 matrix with $a_{ij} = i + j - 1$.
 (c) The 3 by 4 matrix with $a_{ij} = (-1)^j$.

- 3 Find the reduced R for each of these (block) matrices:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 2 & 4 & 6 \end{bmatrix} \quad B = \begin{bmatrix} A & A \end{bmatrix} \quad C = \begin{bmatrix} A & A \\ A & 0 \end{bmatrix}$$

- 4 Suppose all the pivot variables come *last* instead of first. Describe all four blocks in the reduced echelon form (the block B should be r by r):

$$R = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

What is the nullspace matrix N containing the special solutions?

- 5 (Silly problem) Describe all 2 by 3 matrices A_1 and A_2 , with row echelon forms R_1 and R_2 , such that $R_1 + R_2$ is the row echelon form of $A_1 + A_2$. Is it true that $R_1 = A_1$ and $R_2 = A_2$ in this case? Does $R_1 - R_2$ equal $\text{rref}(A_1 - A_2)$?
 6 If A has r pivot columns, how do you know that A^T has r pivot columns? Give a 3 by 3 example with different column numbers in *pivcol* for A and A^T .
 7 What are the special solutions to $Rx = 0$ and $y^T R = 0$ for these R ?

$$R = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Problems 8–11 are about matrices of rank $r = 1$.

- 8 Fill out these matrices so that they have rank 1:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & & \\ 4 & & \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} & 9 & \\ 1 & & \\ 2 & 6 & -3 \end{bmatrix} \quad \text{and} \quad M = \begin{bmatrix} a & b \\ c & \end{bmatrix}.$$

- 9 If A is an m by n matrix with $r = 1$, its columns are multiples of one column and its rows are multiples of one row. The column space is a _____ in $\mathbf{R}^m$. The nullspace is a _____ in $\mathbf{R}^n$. The nullspace matrix N has shape _____.
 10 Choose vectors u and v so that $A = uv^T$ = column times row:

$$A = \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ 4 & 8 & 8 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & 2 & 6 & 4 \\ -1 & -1 & -3 & -2 \end{bmatrix}.$$

$A = uv^T$ is the natural form for every matrix that has rank $r = 1$.

- 11 If A is a rank one matrix, the second row of U is _____. Do an example.

Problems 12–14 are about r by r invertible matrices inside A .

- 12** If A has rank r , then it has an r by r submatrix S that is invertible. Remove $m - r$ rows and $n - r$ columns to find an invertible submatrix S inside A , B , and C . You could keep the pivot rows and pivot columns:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- 13** Suppose P contains only the r pivot columns of an m by n matrix. Explain why this m by r submatrix P has rank r .
- 14** Transpose P in problem 13. Then find the r pivot columns of P^T . Transposing back, **this produces an r by r invertible submatrix S inside P and A :**

$$\text{For } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 4 & 7 \end{bmatrix} \text{ find } P \text{ (3 by 2) and then the invertible } S \text{ (2 by 2).}$$

Problems 15–20 show that $\text{rank}(AB)$ is not greater than $\text{rank}(A)$ or $\text{rank}(B)$.

- 15** Find the ranks of AB and AC (rank one matrix times rank one matrix):

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 1.5 & 6 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & b \\ c & bc \end{bmatrix}.$$

- 16** The rank one matrix uv^T times the rank one matrix wz^T is uz^T times the number _____. This product $uv^T wz^T$ also has rank one unless _____ = 0.
- 17** (a) Suppose column j of B is a combination of previous columns of B . Show that column j of AB is the same combination of previous columns of AB . Then AB cannot have new pivot columns, so **$\text{rank}(AB) \leq \text{rank}(B)$** .
 (b) Find A_1 and A_2 so that $\text{rank}(A_1 B) = 1$ and $\text{rank}(A_2 B) = 0$ for $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
- 18** Problem 17 proved that $\text{rank}(AB) \leq \text{rank}(B)$. Then the same reasoning gives **$\text{rank}(B^T A^T) \leq \text{rank}(A^T)$** . How do you deduce that **$\text{rank}(AB) \leq \text{rank} A$** ?
- 19** (Important) Suppose A and B are n by n matrices, and $AB = I$. Prove from $\text{rank}(AB) \leq \text{rank}(A)$ that the rank of A is n . So A is invertible and B must be its two-sided inverse (Section 2.5). Therefore $BA = I$ (which is not so obvious!).
- 20** If A is 2 by 3 and B is 3 by 2 and $AB = I$, show from its rank that $BA \neq I$. Give an example of A and B with $AB = I$. For $m < n$, a right inverse is not a left inverse.
- 21** Suppose A and B have the same reduced row echelon form R .
 (a) Show that A and B have the same nullspace and the same row space.

(b) We know $E_1 A = R$ and $E_2 B = R$. So A equals an _____ matrix times B .

- 22 Express A and then B as the sum of two rank one matrices:

$$\text{rank} = 2 \quad A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 4 \\ 1 & 1 & 8 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}.$$

- 23 Answer the same questions as in Worked Example 3.3 C for

$$A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 1 & c & 2 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1-c & 2 \\ 0 & 2-c \end{bmatrix}.$$

- 24 What is the nullspace matrix N (containing the special solutions) for A, B, C ?

$$A = \begin{bmatrix} I & I \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} I & I \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} I & I & I \end{bmatrix}.$$

- 25 *Neat fact* Every m by n matrix of rank r reduces to $(m$ by $r)$ times $(r$ by $n)$:

$$A = (\text{pivot columns of } A) (\text{first } r \text{ rows of } R) = (\text{COL})(\text{ROW}).$$

Write the 3 by 4 matrix A in equation (1) at the start of this section as the product of the 3 by 2 matrix from the pivot columns and the 2 by 4 matrix from R .

Challenge Problems

- 26 Suppose A is an m by n matrix of rank r . Its reduced echelon form is R . Describe exactly the matrix Z (its shape and all its entries) that comes from *transposing the reduced row echelon form of R'* (prime means transpose):

$$R = \text{rref}(A) \quad \text{and} \quad Z = (\text{rref}(R'))'.$$

- 27 Suppose R is m by n of rank r , with pivot columns first:

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}.$$

- What are the shapes of those four blocks?
- Find a *right-inverse* B with $RB = I$ if $r = m$.
- Find a *left-inverse* C with $CR = I$ if $r = n$.
- What is the reduced row echelon form of R^T (with shapes)?
- What is the reduced row echelon form of $R^T R$ (with shapes)?

Prove that $R^T R$ has the same nullspace as R . Later we show that $A^T A$ always has the same nullspace as A (a valuable fact).

- 28 Suppose you allow elementary *column* operations on A as well as elementary row operations (which get to R). What is the “row-and-column reduced form” for an m by n matrix of rank r ?

3.4 The Complete Solution to $Ax = b$

The last sections totally solved $Ax = 0$. Elimination converted the problem to $Rx = 0$. The free variables were given special values (one and zero). Then the pivot variables were found by back substitution. We paid no attention to the right side b because it started and ended as zero. The solution x was in the nullspace of A .

Now b is not zero. Row operations on the left side must act also on the right side. $Ax = b$ is reduced to a simpler system $Rx = d$. One way to organize that is to **add b as an extra column of the matrix**. I will “augment” A with the right side $(b_1, b_2, b_3) = (1, 6, 7)$ and reduce the bigger matrix $[A \ b]$:

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 1 & 3 & 1 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 7 \end{bmatrix} \quad \text{has the augmented matrix} \quad \begin{bmatrix} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 1 & 3 & 1 & 6 & 7 \end{bmatrix} = [A \ b].$$

The augmented matrix is just $[A \ b]$. When we apply the usual elimination steps to A , we also apply them to b . That keeps all the equations correct.

In this example we subtract row 1 from row 3 and then subtract row 2 from row 3. This produces a *complete row of zeros* in R , and it changes b to a new right side $d = (1, 6, 0)$:

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} \quad \text{has the augmented matrix} \quad \begin{bmatrix} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = [R \ d].$$

That very last zero is crucial. The third equation has become $0 = 0$ and the equations can be solved. In the original matrix A , the first row plus the second row equals the third row. If the equations are consistent, this must be true on the right side of the equations also! The all-important property on the right side was $1 + 6 = 7$.

Here are the same augmented matrices for a general $b = (b_1, b_2, b_3)$:

$$[A \ b] = \begin{bmatrix} 1 & 3 & 0 & 2 & b_1 \\ 0 & 0 & 1 & 4 & b_2 \\ 1 & 3 & 1 & 6 & b_3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 3 & 0 & 2 & b_1 \\ 0 & 0 & 1 & 4 & b_2 \\ 0 & 0 & 0 & 0 & b_3 - b_1 - b_2 \end{bmatrix} = [R \ d]$$

Now we get $0 = 0$ in the third equation provided $b_3 - b_1 - b_2 = 0$. This is $b_1 + b_2 = b_3$.

One Particular Solution

For an easy solution x , choose the free variables to be $x_2 = x_4 = 0$. Then the two nonzero equations give the two pivot variables $x_1 = 1$ and $x_3 = 6$. Our particular solution to $Ax = b$ (and also $Rx = d$) is $x_p = (1, 0, 6, 0)$. This particular solution is my favorite: *free variables = zero, pivot variables from d* . The method always works.

For a solution to exist, zero rows in R must also be zero in d . Since I is in the pivot rows and pivot columns of R , the pivot variables in $x_{\text{particular}}$ come from d :

$$Rx_p = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} \quad \begin{array}{l} \text{Pivot variables } 1, 6 \\ \text{Free variables } 0, 0 \end{array}$$

Notice how we *choose* the free variables (as zero) and *solve* for the pivot variables. After the row reduction to R , those steps are quick. When the free variables are zero, the pivot variables for x_p are already seen already seen in the right side vector d .

$x_{\text{particular}}$	<i>The particular solution solves</i>	$Ax_p = b$
$x_{\text{nullspace}}$	<i>The $n - r$ special solutions solve</i>	$Ax_n = 0$.

That particular solution is $(1, 0, 6, 0)$. The two special (nullspace) solutions to $Rx = 0$ come from the two free columns of R , by reversing signs of 3, 2, and 4. *Please notice how I write the complete solution $x_p + x_n$ to $Ax = b$:*

Complete solution one x_p many x_n	$x = x_p + x_n = \begin{bmatrix} 1 \\ 0 \\ 6 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ 0 \\ -4 \\ 1 \end{bmatrix}.$
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Question Suppose A is a square invertible matrix, $m = n = r$. What are x_p and x_n ?

Answer The particular solution is the one and *only* solution $A^{-1}b$. There are no special solutions or free variables. $R = I$ has no zero rows. The only vector in the nullspace is $x_n = 0$. The complete solution is $x = x_p + x_n = A^{-1}b + 0$.

This was the situation in Chapter 2. We didn't mention the nullspace in that chapter. $N(A)$ contained only the zero vector. Reduction goes from $[A \ b]$ to $[I \ A^{-1}b]$. The original $Ax = b$ is reduced all the way to $x = A^{-1}b$ which is d . This is a special case here, but square invertible matrices are the ones we see most often in practice. So they got their own chapter at the start of the book.

For small examples we can reduce $[A \ b]$ to $[R \ d]$. For a large matrix, MATLAB does it better. One particular solution (not necessarily ours) is $A \setminus b$ from backslash. Here is an example with *full column rank*. Both columns have pivots.

Example 1 Find the condition on (b_1, b_2, b_3) for $Ax = b$ to be solvable, if

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ -2 & -3 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

This condition puts b in the column space of A . Find the complete $x = x_p + x_n$.

Solution Use the augmented matrix, with its extra column b . Subtract row 1 of $[A \ b]$ from row 2, and add 2 times row 1 to row 3 to reach $[R \ d]$:

$$\begin{bmatrix} 1 & 1 & b_1 \\ 1 & 2 & b_2 \\ -2 & -3 & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & b_1 \\ 0 & 1 & b_2 - b_1 \\ 0 & -1 & b_3 + 2b_1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2b_1 - b_2 \\ 0 & 1 & b_2 - b_1 \\ 0 & 0 & b_3 + b_1 + b_2 \end{bmatrix}.$$

The last equation is $0 = 0$ provided $b_3 + b_1 + b_2 = 0$. This is the condition to put b in the column space; then $Ax = b$ will be solvable. The rows of A add to the zero row. So for consistency (these are equations!) the entries of b must also add to zero.

This example has no free variables since $n - r = 2 - 2$. Therefore no special solutions. The nullspace solution is $x_n = 0$. The particular solution to $Ax = b$ and $Rx = d$ is at the top of the augmented column d :

$$\text{Only solution} \quad x = x_p + x_n = \begin{bmatrix} 2b_1 - b_2 \\ b_2 - b_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

If $b_3 + b_1 + b_2$ is not zero, there is no solution to $Ax = b$ (x_p doesn't exist).

This example is typical of an extremely important case: A has *full column rank*. Every column has a pivot. *The rank is $r = n$* . The matrix is tall and thin ($m \geq n$). Row reduction puts I at the top, when A is reduced to R with rank n :

$$\text{Full column rank} \quad R = \begin{bmatrix} I \\ 0 \end{bmatrix} = \begin{bmatrix} n \text{ by } n \text{ identity matrix} \\ m - n \text{ rows of zeros} \end{bmatrix} \quad (1)$$

There are no free columns or free variables. The nullspace matrix is empty!

We will collect together the different ways of recognizing this type of matrix.

Every matrix A with **full column rank** ($r = n$) has all these properties:

1. All columns of A are pivot columns.
2. There are no free variables or special solutions.
3. The nullspace $N(A)$ contains only the zero vector $x = 0$.
4. If $Ax = b$ has a solution (it might not) then it has only *one solution*.

In the essential language of the next section, **this A has independent columns**. $Ax = 0$ only happens when $x = 0$. In Chapter 4 we will add one more fact to the list: *The square matrix $A^T A$ is invertible when the rank is n .*

In this case the nullspace of A (and R) has shrunk to the zero vector. The solution to $Ax = b$ is *unique* (if it exists). There will be $m - n$ (here $3 - 2$) zero rows in R . So there are $m - n$ conditions in order to have $0 = 0$ in those rows, and b in the column space. With full column rank, $Ax = b$ has *one solution or no solution* ($m > n$ is overdetermined).

The Complete Solution

The other extreme case is full row rank. Now $Ax = b$ has *one or infinitely many* solutions. In this case A must be *short and wide* ($m \leq n$). A matrix has **full row rank** if $r = m$ (“independent rows”). Every row has a pivot, and here is an example.

Example 2 There are $n = 3$ unknowns but only $m = 2$ equations:

$$\begin{array}{lcl} \text{Full row rank} & x + y + z = 3 & \\ & x + 2y - z = 4 & \text{(rank } r = m = 2) \end{array}$$

These are two planes in xyz space. The planes are not parallel so they intersect in a line. This line of solutions is exactly what elimination will find. *The particular solution will be one point on the line. Adding the nullspace vectors x_n will move us along the line.* Then $x = x_p + x_n$ gives the whole line of solutions.

We find x_p and x_n by elimination on $[A \ b]$. Subtract row 1 from row 2 and then subtract row 2 from row 1:

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & -1 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & -2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & -2 & 1 \end{bmatrix} = [R \ d].$$

The particular solution has free variable $x_3 = 0$. The special solution has $x_3 = 1$:

$x_{\text{particular}}$ comes directly from d on the right side: $x_p = (2, 1, 0)$

x_{special} comes from the third column (free column) of R : $s = (-3, 2, 1)$

It is wise to check that x_p and s satisfy the original equations $Ax_p = b$ and $As = 0$:

$$\begin{array}{rcl} 2 + 1 & = & 3 \\ 2 + 2 & = & 4 \end{array} \qquad \begin{array}{rcl} -3 + 2 + 1 & = & 0 \\ -3 + 4 - 1 & = & 0 \end{array}$$

The nullspace solution x_n is any multiple of s . It moves along the line of solutions, starting at $x_{\text{particular}}$. *Please notice again how to write the answer:*

Complete solution

$$x = x_p + x_n = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}.$$

This line is drawn in Figure 3.3. Any point on the line *could* have been chosen as the particular solution; we chose the point with $x_3 = 0$.

The particular solution is *not* multiplied by an arbitrary constant! The special solution is, and you understand why.

Now we summarize this short wide case of *full row rank*. If $m < n$ the equation $Ax = b$ is **underdetermined** (many solutions).

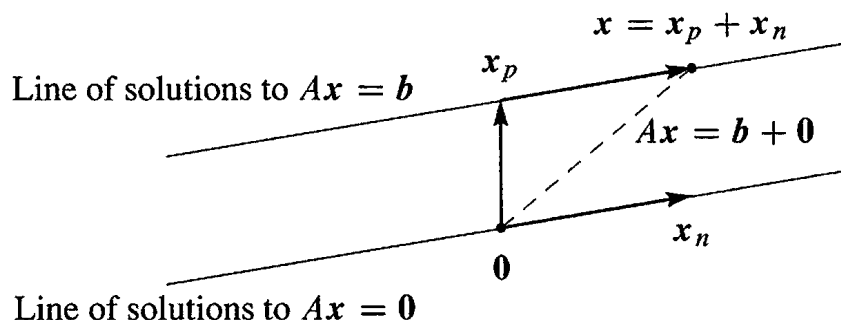


Figure 3.3: Complete solution = *one* particular solution + *all* nullspace solutions.

Every matrix A with **full row rank** ($r = m$) has all these properties:

1. All rows have pivots, and R has no zero rows.
2. $Ax = b$ has a solution for every right side b .
3. The column space is the whole space $\mathbb{R}^m$.
4. There are $n - r = n - m$ special solutions in the nullspace of A .

In this case with m pivots, the rows are “**linearly independent**”. So the columns of A^T are linearly independent. We are more than ready for the definition of linear independence, as soon as we summarize the four possibilities—which depend on the rank. Notice how r , m , n are the critical numbers.

The four possibilities for linear equations depend on the rank r :

$r = m$	and	$r = n$	Square and invertible	$Ax = b$	has 1 solution
$r = m$	and	$r < n$	Short and wide	$Ax = b$	has ∞ solutions
$r < m$	and	$r = n$	Tall and thin	$Ax = b$	has 0 or 1 solution
$r < m$	and	$r < n$	Not full rank	$Ax = b$	has 0 or ∞ solutions

The reduced R will fall in the same category as the matrix A . In case the pivot columns happen to come first, we can display these four possibilities for R . For $Rx = d$ (and the original $Ax = b$) to be solvable, d must end in $m - r$ zeros.

Four types	$R = [I]$	$[I \ F]$	$\begin{bmatrix} I \\ 0 \end{bmatrix}$	$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$
Their ranks	$r = m = n$	$r = m < n$	$r = n < m$	$r < m, r < n$

Cases 1 and 2 have full row rank $r = m$. Cases 1 and 3 have full column rank $r = n$. Case 4 is the most general in theory and it is the least common in practice.

Note My classes used to stop at U before reaching R . Instead of reading the complete solution directly from $Rx = d$, we found it by back substitution from $Ux = c$. This

reduction to U and back substitution for x is slightly faster. Now we prefer the complete reduction: a single “1” in each pivot column. Everything is so clear in R (and the computer should do the hard work anyway) that we reduce all the way.

■ REVIEW OF THE KEY IDEAS ■

1. The rank r is the number of pivots. The matrix R has $m - r$ zero rows.
2. $Ax = b$ is solvable if and only if the last $m - r$ equations reduce to $0 = 0$.
3. One particular solution x_p has all free variables equal to zero.
4. The pivot variables are determined after the free variables are chosen.
5. Full column rank $r = n$ means no free variables: one solution or none.
6. Full row rank $r = m$ means one solution if $m = n$ or infinitely many if $m < n$.

■ WORKED EXAMPLES ■

3.4 A This question connects elimination (**pivot columns and back substitution**) to **column space-nullspace-rank-solvability** (the full picture). A has rank 2:

$$Ax = b \quad \text{is} \quad \begin{aligned} x_1 + 2x_2 + 3x_3 + 5x_4 &= b_1 \\ 2x_1 + 4x_2 + 8x_3 + 12x_4 &= b_2 \\ 3x_1 + 6x_2 + 7x_3 + 13x_4 &= b_3 \end{aligned}$$

1. Reduce $[A \ b]$ to $[U \ c]$, so that $Ax = b$ becomes a triangular system $Ux = c$.
2. Find the condition on b_1, b_2, b_3 for $Ax = b$ to have a solution.
3. Describe the column space of A . Which plane in $\mathbf{R}^3$?
4. Describe the nullspace of A . Which special solutions in $\mathbf{R}^4$?
5. Find a particular solution to $Ax = (0, 6, -6)$ and then the complete solution.
6. Reduce $[U \ c]$ to $[R \ d]$: Special solutions from R , particular solution from d .

Solution

1. The multipliers in elimination are 2 and 3 and -1 . They take $[A \ b]$ into $[U \ c]$.

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & \mathbf{b}_1 \\ 2 & 4 & 8 & 12 & \mathbf{b}_2 \\ 3 & 6 & 7 & 13 & \mathbf{b}_3 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & \mathbf{b}_1 \\ 0 & 0 & 2 & 2 & \mathbf{b}_2 - 2\mathbf{b}_1 \\ 0 & 0 & -2 & -2 & \mathbf{b}_3 - 3\mathbf{b}_1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & \mathbf{b}_1 \\ 0 & 0 & 2 & 2 & \mathbf{b}_2 - 2\mathbf{b}_1 \\ 0 & 0 & 0 & 0 & \mathbf{b}_3 + \mathbf{b}_2 - 5\mathbf{b}_1 \end{array} \right]$$

2. The last equation shows the solvability condition $b_3 + b_2 - 5b_1 = 0$. Then $0 = 0$.
3. **First description:** The column space is the plane containing all combinations of the pivot columns (1, 2, 3) and (3, 8, 7). The pivots are in columns 1 and 3. **Second description:** The column space contains all vectors with $b_3 + b_2 - 5b_1 = 0$. That makes $Ax = b$ solvable, so b is in the column space. *All columns of A pass this test $b_3 + b_2 - 5b_1 = 0$. This is the equation for the plane in the first description!*
4. The special solutions have free variables $x_2 = 1, x_4 = 0$ and then $x_2 = 0, x_4 = 1$:

$$\begin{array}{ll} \text{Special solutions to } Ax = 0 & s_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad s_2 = \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix} \\ \text{Back substitution in } Ux = 0 & \end{array}$$

The nullspace $N(A)$ in $\mathbb{R}^4$ contains all $x_n = c_1 s_1 + c_2 s_2$.

5. One particular solution x_p has free variables = zero. Back substitute in $Ux = c$:

$$\begin{array}{ll} \text{Particular solution to } Ax_p = b = (0, 6, -6) & x_p = \begin{bmatrix} -9 \\ 0 \\ 3 \\ 0 \end{bmatrix} \\ \text{This vector } b \text{ satisfies } b_3 + b_2 - 5b_1 = 0 & \end{array}$$

The complete solution to $Ax = (0, 6, -6)$ is $x = x_p + \text{all } x_n$.

6. In the reduced form R , the third column changes from (3, 2, 0) in U to (0, 1, 0). The right side $c = (0, 6, 0)$ becomes $d = (-9, 3, 0)$ showing -9 and 3 in x_p :

$$[U \quad c] = \begin{bmatrix} 1 & 2 & 3 & 5 & 0 \\ 0 & 0 & 2 & 2 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow [R \quad d] = \begin{bmatrix} 1 & 2 & 0 & 2 & -9 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

3.4 B If you have this information about the solutions to $Ax = b$ for a specific b , what does that tell you about the shape of A (and A itself)? And possibly about b .

1. There is exactly one solution.
2. All solutions to $Ax = b$ have the form $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
3. There are no solutions.
4. All solutions to $Ax = b$ have the form $x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$
5. There are infinitely many solutions.

Solution In case 1, with exactly one solution, A must have full column rank $r = n$. The nullspace of A contains only the zero vector. Necessarily $m \geq n$.

In case 2, A must have $n = 2$ columns (and m is arbitrary). With $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ in the nullspace of A , column 2 is the *negative* of column 1. Also $A \neq 0$: the rank is 1. With $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ as a solution, $b = 2(\text{column 1}) + (\text{column 2})$. My choice for x_p would be (1, 0).

In case 3 we only know that b is not in the column space of A . The rank of A must be less than m . I guess we know $b \neq 0$, otherwise $x = 0$ would be a solution.

In case 4, A must have $n = 3$ columns. With $(1, 0, 1)$ in the nullspace of A , column 3 is the negative of column 1. Column 2 must *not* be a multiple of column 1, or the nullspace would contain another special solution. So the rank of A is $3 - 1 = 2$. Necessarily A has $m \geq 2$ rows. The right side $\mathbf{b}$ is column 1 + column 2.

In case 5 with infinitely many solutions, the nullspace must contain nonzero vectors. The rank r must be less than n (not full column rank), and $\mathbf{b}$ must be in the column space of A . We don't know if *every* $\mathbf{b}$ is in the column space, so we don't know if $r = m$.

3.4 C Find the complete solution $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ by forward elimination on $[A \ \mathbf{b}]$:

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 4 & 8 \\ 4 & 8 & 6 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 10 \end{bmatrix}.$$

Find numbers y_1, y_2, y_3 so that y_1 (row 1) + y_2 (row 2) + y_3 (row 3) = **zero row**. Check that $\mathbf{b} = (4, 2, 10)$ satisfies the condition $y_1 b_1 + y_2 b_2 + y_3 b_3 = 0$. Why is this the condition for the equations to be solvable and $\mathbf{b}$ to be in the column space?

Solution Forward elimination on $[A \ \mathbf{b}]$ produces a zero row in $[U \ \mathbf{c}]$. The third equation becomes $0 = 0$ and the equations are consistent (and solvable):

$$\begin{bmatrix} 1 & 2 & 1 & 0 & 4 \\ 2 & 4 & 4 & 8 & 2 \\ 4 & 8 & 6 & 8 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 2 & 8 & -6 \\ 0 & 0 & 2 & 8 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 2 & 8 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 and 3 contain pivots. The variables x_2 and x_4 are free. If we set those to zero we can solve (back substitution) for the particular solution $\mathbf{x}_p = (7, 0, -3, 0)$. We see 7 and -3 again if elimination continues all the way to $[R \ \mathbf{d}]$:

$$\begin{bmatrix} 1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 2 & 8 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & -4 & 7 \\ 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

For the nullspace part $\mathbf{x}_n$ with $\mathbf{b} = \mathbf{0}$, set the free variables x_2, x_4 to 1, 0 and also 0, 1:

Special solutions $\mathbf{s}_1 = (-2, 1, 0, 0)$ and $\mathbf{s}_2 = (4, 0, -4, 1)$

Then the complete solution to $A\mathbf{x} = \mathbf{b}$ (and $R\mathbf{x} = \mathbf{d}$) is $\mathbf{x}_{\text{complete}} = \mathbf{x}_p + c_1 \mathbf{s}_1 + c_2 \mathbf{s}_2$.

The rows of A produced the zero row from $2(\text{row } 1) + (\text{row } 2) - (\text{row } 3) = (0, 0, 0, 0)$. Thus $\mathbf{y} = (2, 1, -1)$. The same combination for $\mathbf{b} = (4, 2, 10)$ gives $2(4) + (2) - (10) = 0$.

If a combination of the rows (on the left side) gives the zero row, then the same combination must give zero on the right side. Of course! *Otherwise no solution.*

Later we will say this again in different words: If every column of A is perpendicular to $\mathbf{y} = (2, 1, -1)$, then any combination $\mathbf{b}$ of those columns must also be perpendicular to $\mathbf{y}$. Otherwise $\mathbf{b}$ is not in the column space and $A\mathbf{x} = \mathbf{b}$ is not solvable.

And again: If $\mathbf{y}$ is in the nullspace of A^T then $\mathbf{y}$ must be perpendicular to every $\mathbf{b}$ in the column space of A . Just looking ahead...

Problem Set 3.4

- 1 (Recommended) Execute the six steps of Worked Example 3.4 A to describe the column space and nullspace of A and the complete solution to $Ax = b$:

$$A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

- 2 Carry out the same six steps for this matrix A with rank one. You will find *two* conditions on b_1, b_2, b_3 for $Ax = b$ to be solvable. Together these two conditions put b into the _____ space (two planes give a line):

$$A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} [2 \quad 1 \quad 3] = \begin{bmatrix} 2 & 1 & 3 \\ 6 & 3 & 9 \\ 4 & 2 & 6 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 30 \\ 20 \end{bmatrix}$$

Questions 3–15 are about the solution of $Ax = b$. Follow the steps in the text to x_p and x_n . Use the augmented matrix with last column b .

- 3 Write the complete solution as x_p plus any multiple of s in the nullspace:

$$\begin{aligned} x + 3y + 3z &= 1 \\ 2x + 6y + 9z &= 5 \\ -x - 3y + 3z &= 5. \end{aligned}$$

- 4 Find the complete solution (also called the *general solution*) to

$$\begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

- 5 Under what condition on b_1, b_2, b_3 is this system solvable? Include b as a fourth column in elimination. Find all solutions when that condition holds:

$$\begin{aligned} x + 2y - 2z &= b_1 \\ 2x + 5y - 4z &= b_2 \\ 4x + 9y - 8z &= b_3. \end{aligned}$$

- 6 What conditions on b_1, b_2, b_3, b_4 make each system solvable? Find x in that case:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 2 & 5 \\ 3 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 5 & 7 \\ 3 & 9 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}.$$

- 7 Show by elimination that (b_1, b_2, b_3) is in the column space if $b_3 - 2b_2 + 4b_1 = 0$.

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 8 & 2 \\ 2 & 4 & 0 \end{bmatrix}.$$

What combination of the rows of A gives the zero row?

- 8 Which vectors (b_1, b_2, b_3) are in the column space of A ? Which combinations of the rows of A give zero?

(a) $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 6 & 3 \\ 0 & 2 & 5 \end{bmatrix}$

(b) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix}.$

- 9 (a) The Worked Example 3.4 A reached $[U \ c]$ from $[A \ b]$. Put the multipliers into L and verify that LU equals A and Lc equals b .
 (b) Combine the pivot columns of A with the numbers -9 and 3 in the particular solution x_p . What is that linear combination and why?
- 10 Construct a 2 by 3 system $Ax = b$ with particular solution $x_p = (2, 4, 0)$ and homogeneous solution $x_n =$ any multiple of $(1, 1, 1)$.
- 11 Why can't a 1 by 3 system have $x_p = (2, 4, 0)$ and $x_n =$ any multiple of $(1, 1, 1)$?
- 12 (a) If $Ax = b$ has two solutions x_1 and x_2 , find two solutions to $Ax = 0$.
 (b) Then find another solution to $Ax = 0$ and another solution to $Ax = b$.
- 13 Explain why these are all false:
 (a) The complete solution is any linear combination of x_p and x_n .
 (b) A system $Ax = b$ has at most one particular solution.
 (c) The solution x_p with all free variables zero is the shortest solution (minimum length $\|x\|$). Find a 2 by 2 counterexample.
 (d) If A is invertible there is no solution x_n in the nullspace.
- 14 Suppose column 5 of U has no pivot. Then x_5 is a _____ variable. The zero vector (is) (is not) the only solution to $Ax = 0$. If $Ax = b$ has a solution, then it has _____ solutions.
- 15 Suppose row 3 of U has no pivot. Then that row is _____. The equation $Ux = c$ is only solvable provided _____. The equation $Ax = b$ (is) (is not) (might not be) solvable.

Questions 16–20 are about matrices of “full rank” $r = m$ or $r = n$.

- 16 The largest possible rank of a 3 by 5 matrix is _____. Then there is a pivot in every _____ of U and R . The solution to $Ax = b$ (always exists) (is unique). The column space of A is _____. An example is $A =$ _____.

- 17 The largest possible rank of a 6 by 4 matrix is _____. Then there is a pivot in every _____ of U and R . The solution to $Ax = b$ (*always exists*) (*is unique*). The nullspace of A is _____. An example is $A =$ _____.

- 18 Find by elimination the rank of A and also the rank of A^T :

$$A = \begin{bmatrix} 1 & 4 & 0 \\ 2 & 11 & 5 \\ -1 & 2 & 10 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & q \end{bmatrix} \quad (\text{rank depends on } q).$$

- 19 Find the rank of A and also of $A^T A$ and also of AA^T :

$$A = \begin{bmatrix} 1 & 1 & 5 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

- 20 Reduce A to its echelon form U . Then find a triangular L so that $A = LU$.

$$A = \begin{bmatrix} 3 & 4 & 1 & 0 \\ 6 & 5 & 2 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 3 \\ 0 & 6 & 5 & 4 \end{bmatrix}.$$

- 21 Find the complete solution in the form $x_p + x_n$ to these full rank systems:

$$\begin{array}{ll} \text{(a)} & x + y + z = 4 \\ \text{(b)} & \begin{array}{l} x + y + z = 4 \\ x - y + z = 4. \end{array} \end{array}$$

- 22 If $Ax = b$ has infinitely many solutions, why is it impossible for $Ax = B$ (new right side) to have only one solution? Could $Ax = B$ have no solution?

- 23 Choose the number q so that (if possible) the ranks are (a) 1, (b) 2, (c) 3:

$$A = \begin{bmatrix} 6 & 4 & 2 \\ -3 & -2 & -1 \\ 9 & 6 & q \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 1 & 3 \\ q & 2 & q \end{bmatrix}.$$

- 24 Give examples of matrices A for which the number of solutions to $Ax = b$ is

- (a) 0 or 1, depending on b
- (b) ∞ , regardless of b
- (c) 0 or ∞ , depending on b
- (d) 1, regardless of b .

- 25 Write down all known relations between r and m and n if $Ax = b$ has

- (a) no solution for some b

- (b) infinitely many solutions for every b
- (c) exactly one solution for some b , no solution for other b
- (d) exactly one solution for every b .

Questions 26–33 are about Gauss-Jordan elimination (upwards as well as downwards) and the reduced echelon matrix R .

- 26** Continue elimination from U to R . Divide rows by pivots so the new pivots are all 1. Then produce zeros *above* those pivots to reach R :

$$U = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 2 & 4 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 5 \end{bmatrix}.$$

- 27** Suppose U is square with n pivots (an invertible matrix). *Explain why $R = I$.*
- 28** Apply Gauss-Jordan elimination to $Ux = 0$ and $Ux = c$. Reach $Rx = 0$ and $Rx = d$:

$$[U \ 0] = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix} \quad \text{and} \quad [U \ c] = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \end{bmatrix}.$$

Solve $Rx = 0$ to find x_n (its free variable is $x_2 = 1$). Solve $Rx = d$ to find x_p (its free variable is $x_2 = 0$).

- 29** Apply Gauss-Jordan elimination to reduce to $Rx = 0$ and $Rx = d$:

$$\begin{bmatrix} U & 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 6 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} U & c \end{bmatrix} = \begin{bmatrix} 3 & 0 & 6 & 9 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

Solve $Ux = 0$ or $Rx = 0$ to find x_n (free variable = 1). What are the solutions to $Rx = d$?

- 30** Reduce to $Ux = c$ (Gaussian elimination) and then $Rx = d$ (Gauss-Jordan):

$$Ax = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 \\ 2 & 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 10 \end{bmatrix} = b.$$

Find a particular solution x_p and all homogeneous solutions x_n .

- 31** Find matrices A and B with the given property or explain why you can't:

- (a) The only solution of $Ax = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is $x = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.
- (b) The only solution of $Bx = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

- 32** Find the LU factorization of A and the complete solution to $Ax = b$:

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 5 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 5 \end{bmatrix} \quad \text{and then} \quad b = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

- 33** The complete solution to $Ax = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Find A .

Challenge Problems

- 34** Suppose you know that the 3 by 4 matrix A has the vector $s = (2, 3, 1, 0)$ as the only special solution to $Ax = 0$.
- (a) What is the *rank* of A and the complete solution to $Ax = 0$?
 - (b) What is the exact row reduced echelon form R of A ?
 - (c) How do you know that $Ax = b$ can be solved for all b ?
- 35** Suppose K is the 9 by 9 second difference matrix (2's on the diagonal, -1 's on the diagonal above and also below). Solve the equation $Kx = b = (10, \dots, 10)$. If you graph $x_1, \dots, x_9$ above the points $1, \dots, 9$ on the x axis, I think the nine points fall on a parabola.
- 36** Suppose $Ax = b$ and $Cx = b$ have the same (complete) solutions for every b . Is it true that $A = C$?

3.5 Independence, Basis and Dimension

This important section is about the true size of a subspace. There are n columns in an m by n matrix. But the true “dimension” of the column space is not necessarily n . The dimension is measured by counting *independent columns*—and we have to say what that means. We will see that *the true dimension of the column space is the rank r* .

The idea of independence applies to any vectors $v_1, \dots, v_n$ in any vector space. Most of this section concentrates on the subspaces that we know and use—especially the column space and the nullspace of A . In the last part we also study “vectors” that are not column vectors. They can be matrices and functions; they can be linearly independent (or dependent). First come the key examples using column vectors.

The goal is to understand a **basis**: **independent vectors that “span the space”**.

Every vector in the space is a unique combination of the basis vectors.

We are at the heart of our subject, and we cannot go on without a basis. The four essential ideas in this section (with first hints at their meaning) are:

- | | |
|-------------------------|--------------------------------------|
| 1. Independent vectors | (no extra vectors) |
| 2. Spanning a space | (enough vectors to produce the rest) |
| 3. Basis for a space | (not too many or too few) |
| 4. Dimension of a space | (the number of vectors in a basis) |

Linear Independence

Our first definition of independence is not so conventional, but you are ready for it.

DEFINITION The columns of A are *linearly independent* when the only solution to $Ax = 0$ is $x = 0$. *No other combination Ax of the columns gives the zero vector.*

The columns are independent when the nullspace $N(A)$ contains only the zero vector. Let me illustrate linear independence (and dependence) with three vectors in $\mathbf{R}^3$:

1. If three vectors are *not* in the same plane, they are independent. No combination of v_1, v_2, v_3 in Figure 3.4 gives zero except $0v_1 + 0v_2 + 0v_3$.
2. If three vectors w_1, w_2, w_3 are *in the same plane*, they are dependent.

This idea of independence applies to 7 vectors in 12-dimensional space. If they are the columns of A , and independent, the nullspace only contains $x = 0$. None of the vectors is a combination of the other six vectors.

Now we choose different words to express the same idea. The following definition of independence will apply to any sequence of vectors in any vector space. When the vectors are the columns of A , the two definitions say exactly the same thing.

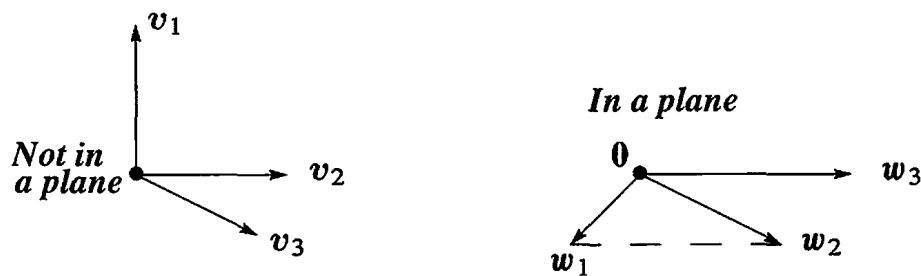


Figure 3.4: Independent vectors v_1, v_2, v_3 . Only $0v_1 + 0v_2 + 0v_3$ gives the vector $\mathbf{0}$. Dependent vectors w_1, w_2, w_3 . The combination $w_1 - w_2 + w_3$ is $(0, 0, 0)$.

DEFINITION The sequence of vectors $v_1, \dots, v_n$ is *linearly independent* if the only combination that gives the zero vector is $0v_1 + 0v_2 + \dots + 0v_n$.

Linear independence

$$x_1v_1 + x_2v_2 + \dots + x_nv_n = \mathbf{0} \quad \text{only happens when all } x\text{'s are zero.} \quad (1)$$

If a combination gives $\mathbf{0}$, when the x 's are not all zero, the vectors are *dependent*.

Correct language: "The sequence of vectors is linearly independent." *Acceptable shortcut:* "The vectors are independent." *Unacceptable:* "The matrix is independent."

A sequence of vectors is either dependent or independent. They can be combined to give the zero vector (with nonzero x 's) or they can't. So the key question is: Which combinations of the vectors give zero? We begin with some small examples in $\mathbf{R}^2$:

- (a) The vectors $(1, 0)$ and $(0, 1)$ are independent.
- (b) The vectors $(1, 0)$ and $(1, 0.00001)$ are independent.
- (c) The vectors $(1, 1)$ and $(-1, -1)$ are *dependent*.
- (d) The vectors $(1, 1)$ and $(0, 0)$ are *dependent* because of the zero vector.
- (e) In $\mathbf{R}^2$, any three vectors (a, b) and (c, d) and (e, f) are *dependent*.

Geometrically, $(1, 1)$ and $(-1, -1)$ are on a line through the origin. They are dependent. To use the definition, find numbers x_1 and x_2 so that $x_1(1, 1) + x_2(-1, -1) = (0, 0)$. This is the same as solving $Ax = \mathbf{0}$:

$$\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{for } x_1 = 1 \text{ and } x_2 = 1.$$

The columns are dependent exactly when *there is a nonzero vector in the nullspace*.

If one of the v 's is the zero vector, independence has no chance. Why not?

Three vectors in $\mathbf{R}^2$ cannot be independent! One way to see this: the matrix A with those three columns must have a free variable and then a special solution to $Ax = \mathbf{0}$. Another way: If the first two vectors are independent, some combination will produce the third vector. See the second highlight below.

Now move to three vectors in $\mathbf{R}^3$. If one of them is a multiple of another one, these vectors are dependent. But the complete test involves all three vectors at once. We put them in a matrix and try to solve $Ax = \mathbf{0}$.

Example 1 The columns of this A are dependent. $Ax = \mathbf{0}$ has a nonzero solution:

$$Ax = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 5 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \text{ is } -3 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 3 \\ 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The rank is only $r = 2$. *Independent columns produce full column rank $r = n = 3$.*

In that matrix the rows are also dependent. Row 1 minus row 3 is the zero row. For a square matrix, we will show that dependent columns imply dependent rows.

Question How to find that solution to $Ax = \mathbf{0}$? The systematic way is elimination.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 5 \\ 1 & 0 & 3 \end{bmatrix} \text{ reduces to } R = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The solution $x = (-3, 1, 1)$ was exactly the special solution. It shows how the free column (column 3) is a combination of the pivot columns. That kills independence!

Full column rank The columns of A are independent exactly when the rank is $r = n$. There are n pivots and no free variables. Only $x = \mathbf{0}$ is in the nullspace.

One case is of special importance because it is clear from the start. Suppose seven columns have five components each ($m = 5$ is less than $n = 7$). Then the columns *must be dependent*. Any seven vectors from $\mathbf{R}^5$ are dependent. The rank of A cannot be larger than 5. There cannot be more than five pivots in five rows. $Ax = \mathbf{0}$ has at least $7 - 5 = 2$ free variables, so it has nonzero solutions—which means that the columns are dependent.

Any set of n vectors in $\mathbf{R}^m$ must be linearly dependent if $n > m$.

This type of matrix has more columns than rows—it is short and wide. The columns are certainly dependent if $n > m$, because $Ax = \mathbf{0}$ has a nonzero solution.

The columns might be dependent or might be independent if $n \leq m$. Elimination will reveal the r pivot columns. *It is those r pivot columns that are independent.*

Note Another way to describe linear dependence is this: “One vector is a combination of the other vectors.” That sounds clear. Why don’t we say this from the start? Our definition was longer: “Some combination gives the zero vector, other than the trivial combination with every $x = 0$.” We must rule out the easy way to get the zero vector.

That trivial combination of zeros gives every author a headache. If one vector is a combination of the others, that vector has coefficient $x = 1$.

The point is, our definition doesn't pick out one particular vector as guilty. All columns of A are treated the same. We look at $Ax = 0$, and it has a nonzero solution or it hasn't. In the end that is better than asking if the last column (or the first, or a column in the middle) is a combination of the others.

Vectors that Span a Subspace

The first subspace in this book was the column space. Starting with columns $v_1, \dots, v_n$, the subspace was filled out by including all combinations $x_1 v_1 + \dots + x_n v_n$. *The column space consists of all combinations Ax of the columns.* We now introduce the single word "span" to describe this: The column space is *spanned* by the columns.

DEFINITION A set of vectors *spans* a space if their linear combinations fill the space.

The columns of a matrix span its column space. They might be dependent.

Example 2 $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ span the full two-dimensional space $\mathbf{R}^2$.

Example 3 $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$ also span the full space $\mathbf{R}^2$.

Example 4 $w_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $w_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ only span a line in $\mathbf{R}^2$. So does w_1 by itself.

Think of two vectors coming out from $(0, 0, 0)$ in 3-dimensional space. Generally they span a plane. Your mind fills in that plane by taking linear combinations. Mathematically you know other possibilities: two vectors could span a line, three vectors could span all of $\mathbf{R}^3$, or only a plane. It is even possible that three vectors span only a line, or ten vectors span only a plane. They are certainly not independent!

The columns span the column space. Here is a new subspace—which is *spanned by the rows*. *The combinations of the rows produce the "row space".*

DEFINITION The *row space* of a matrix is the subspace of $\mathbf{R}^n$ spanned by the rows.

The row space of A is $C(A^T)$. It is the column space of A^T .

The rows of an m by n matrix have n components. They are vectors in $\mathbf{R}^n$ —or they would be if they were written as column vectors. There is a quick way to fix that: *Transpose the matrix*. Instead of the rows of A , look at the columns of A^T . Same numbers, but now in the column space $C(A^T)$. This row space of A is a subspace of $\mathbf{R}^n$.

Example 5 Describe the column space and the row space of A .

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 7 \\ 3 & 5 \end{bmatrix} \text{ and } A^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 7 & 5 \end{bmatrix}. \text{ Here } m = 3 \text{ and } n = 2.$$

The column space of A is the plane in $\mathbf{R}^3$ spanned by the two columns of A . *The row space of A is spanned by the three rows of A* (which are columns of A^T). This row space is all of $\mathbf{R}^2$. Remember: The rows are in $\mathbf{R}^n$ spanning the row space. The columns are in $\mathbf{R}^m$ spanning the column space. Same numbers, different vectors, different spaces.

A Basis for a Vector Space

Two vectors can't span all of $\mathbf{R}^3$, even if they are independent. Four vectors can't be independent, even if they span $\mathbf{R}^3$. We want *enough independent vectors to span the space* (and not more). A “*basis*” is just right.

DEFINITION A *basis* for a vector space is a sequence of vectors with two properties:

The basis vectors are linearly independent and they span the space.

This combination of properties is fundamental to linear algebra. Every vector v in the space is a combination of the basis vectors, because they span the space. More than that, the combination that produces v is *unique*, because the basis vectors $v_1, \dots, v_n$ are independent:

There is one and only one way to write v as a combination of the basis vectors.

Reason: Suppose $v = a_1v_1 + \dots + a_nv_n$ and also $v = b_1v_1 + \dots + b_nv_n$. By subtraction $(a_1 - b_1)v_1 + \dots + (a_n - b_n)v_n$ is the zero vector. From the independence of the v 's, each $a_i - b_i = 0$. Hence $a_i = b_i$, and there are not two ways to produce v .

Example 6 The columns of $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ produce the “standard basis” for $\mathbf{R}^2$.

The basis vectors $i = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $j = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are independent. They span $\mathbf{R}^2$.

Everybody thinks of this basis first. The vector i goes across and j goes straight up. The columns of the 3 by 3 identity matrix are the standard basis i, j, k . The columns of the n by n identity matrix give the “standard basis” for $\mathbf{R}^n$.

Now we find many other bases (infinitely many). The basis is not unique!

Example 7 (Important) The columns of *every invertible n by n matrix* give a basis for $\mathbf{R}^n$:

Invertible matrix	$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$	Singular matrix	$B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix}$
Independent columns		Dependent columns	
Column space is $\mathbf{R}^3$		Column space $\neq \mathbf{R}^3$	

The only solution to $Ax = \mathbf{0}$ is $x = A^{-1}\mathbf{0} = \mathbf{0}$. The columns are independent. They span the whole space $\mathbf{R}^n$ —because every vector b is a combination of the columns. $Ax = b$ can always be solved by $x = A^{-1}b$. Do you see how everything comes together for invertible matrices? Here it is in one sentence:

The vectors $v_1, \dots, v_n$ are a *basis for $\mathbf{R}^n$* exactly when they are *the columns of an n by n invertible matrix*. Thus $\mathbf{R}^n$ has infinitely many different bases.

When the columns are dependent, we keep only the *pivot columns*—the first two columns of B above, with its two pivots. They are independent and they span the column space.

The pivot columns of A are a basis for its column space. The pivot rows of A are a basis for its row space. So are the pivot rows of its echelon form R .

Example 8 This matrix is not invertible. Its columns are not a basis for anything!

One pivot column
One pivot row ($r = 1$)

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix} \text{ reduces to } R = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

Column 1 of A is the pivot column. That column alone is a basis for its column space. The second column of A would be a different basis. So would any nonzero multiple of that column. There is no shortage of bases. One definite choice is the pivot columns.

Notice that the pivot column $(1, 0)$ of this R ends in zero. That column is a basis for the column space of R , but it doesn't belong to the column space of A . The column spaces of A and R are different. Their bases are different. (Their dimensions are the same.)

The row space of A is the *same* as the row space of R . It contains $(2, 4)$ and $(1, 2)$ and all other multiples of those vectors. As always, there are infinitely many bases to choose from. One natural choice is to pick the nonzero rows of R (rows with a pivot). So this matrix A with rank one has only one vector in the basis:

$$\text{Basis for the column space: } \begin{bmatrix} 2 \\ 3 \end{bmatrix}. \quad \text{Basis for the row space: } \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The next chapter will come back to these bases for the column space and row space. We are happy first with examples where the situation is clear (and the idea of a basis is still new). The next example is larger but still clear.

Example 9 Find bases for the column and row spaces of this rank two matrix:

$$R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 and 3 are the pivot columns. They are a basis for the column space (of R !). The vectors in that column space all have the form $\mathbf{b} = (x, y, 0)$. The column space of R is the “ xy plane” inside the full 3-dimensional xyz space. That plane is not $\mathbf{R}^2$, it is a subspace of $\mathbf{R}^3$. Columns 2 and 3 are also a basis for the same column space. Which pairs of columns of R are *not* a basis for its column space?

The row space of R is a subspace of $\mathbf{R}^4$. The simplest basis for that row space is the two nonzero rows of R . The third row (the zero vector) is in the row space too. But it is not in a *basis* for the row space. The basis vectors must be independent.

Question Given five vectors in $\mathbf{R}^7$, *how do you find a basis for the space they span?*

First answer Make them the rows of A , and eliminate to find the nonzero rows of R .

Second answer Put the five vectors into the columns of A . Eliminate to find the pivot columns (of A not R). The program `colbasis` uses the column numbers from `pivcol`.

Could another basis have more vectors, or fewer? This is a crucial question with a good answer: *No. All bases for a vector space contain the same number of vectors.*

The number of vectors, in any and every basis, is the “dimension” of the space.

Dimension of a Vector Space

We have to prove what was just stated. There are many choices for the basis vectors, but the *number* of basis vectors doesn’t change.

If $\mathbf{v}_1, \dots, \mathbf{v}_m$ and $\mathbf{w}_1, \dots, \mathbf{w}_n$ are both bases for the same vector space, then $m = n$.

Proof Suppose that there are more $\mathbf{w}$ ’s than $\mathbf{v}$ ’s. From $n > m$ we want to reach a contradiction. The $\mathbf{v}$ ’s are a basis, so $\mathbf{w}_1$ must be a combination of the $\mathbf{v}$ ’s. If $\mathbf{w}_1$ equals $a_{11}\mathbf{v}_1 + \dots + a_{m1}\mathbf{v}_m$, this is the first column of a matrix multiplication VA :

$$\begin{array}{l} \text{Each } \mathbf{w} \text{ is a} \\ \text{combination} \\ \text{of the } \mathbf{v}\text{'s} \end{array} \quad W = \begin{bmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \dots & \mathbf{w}_n \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_m \end{bmatrix} \begin{bmatrix} a_{11} & a_{1n} \\ \vdots & \vdots \\ a_{m1} & a_{mn} \end{bmatrix} = VA.$$

We don’t know each a_{ij} , but we know the shape of A (it is m by n). The second vector $\mathbf{w}_2$ is also a combination of the $\mathbf{v}$ ’s. The coefficients in that combination fill the second column of A . The key is that A has a row for every $\mathbf{v}$ and a column for every $\mathbf{w}$. A is a short wide matrix, since we assumed $n > m$. So $A\mathbf{x} = \mathbf{0}$ *has a nonzero solution*.

$A\mathbf{x} = \mathbf{0}$ gives $VA\mathbf{x} = \mathbf{0}$ which is $W\mathbf{x} = \mathbf{0}$. A combination of the $\mathbf{w}$ ’s gives zero! Then the $\mathbf{w}$ ’s could not be a basis—our assumption $n > m$ is **not possible** for two bases.

If $m > n$ we exchange the $\mathbf{v}$ ’s and $\mathbf{w}$ ’s and repeat the same steps. The only way to avoid a contradiction is to have $m = n$. This completes the proof that $m = n$.

The number of basis vectors depends on the space—not on a particular basis. The number is the same for every basis, and it counts the “degrees of freedom” in the space.

The dimension of the space $\mathbf{R}^n$ is n . We now introduce the important word *dimension* for other vector spaces too.

DEFINITION The *dimension of a space* is the *number of vectors* in every basis.

This matches our intuition. The line through $v = (1, 5, 2)$ has dimension one. It is a subspace with this one vector v in its basis. Perpendicular to that line is the plane $x + 5y + 2z = 0$. This plane has dimension 2. To prove it, we find a basis $(-5, 1, 0)$ and $(-2, 0, 1)$. The dimension is 2 because the basis contains two vectors.

The plane is the nullspace of the matrix $A = \begin{bmatrix} 1 & 5 & 2 \end{bmatrix}$, which has two free variables. Our basis vectors $(-5, 1, 0)$ and $(-2, 0, 1)$ are the “special solutions” to $Ax = 0$. The next section shows that the $n - r$ special solutions always give a basis for the nullspace. $C(A)$ has dimension r and the nullspace $N(A)$ has dimension $n - r$.

Note about the language of linear algebra We never say “the rank of a space” or “the dimension of a basis” or “the basis of a matrix”. Those terms have no meaning. It is the *dimension of the column space* that equals the *rank of the matrix*.

Bases for Matrix Spaces and Function Spaces

The words “independence” and “basis” and “dimension” are not at all restricted to column vectors. We can ask whether three matrices A_1, A_2, A_3 are independent. When they are in the space of all 3 by 4 matrices, some combination might give the zero matrix. We can also ask the dimension of the full 3 by 4 matrix space. (It is 12.)

In differential equations, $d^2y/dx^2 = y$ has a space of solutions. One basis is $y = e^x$ and $y = e^{-x}$. Counting the basis functions gives the dimension 2 for the space of all solutions. (The dimension is 2 because of the second derivative.)

Matrix spaces and function spaces may look a little strange after $\mathbf{R}^n$. But in some way, you haven’t got the ideas of basis and dimension straight until you can apply them to “vectors” other than column vectors.

Matrix spaces The vector space $\mathbf{M}$ contains all 2 by 2 matrices. Its dimension is 4.

$$\text{One basis is } A_1, A_2, A_3, A_4 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Those matrices are linearly independent. We are not looking at their columns, but at the whole matrix. Combinations of those four matrices can produce any matrix in $\mathbf{M}$, so they span the space:

$$\text{Every } A \text{ combines the basis matrices } c_1 A_1 + c_2 A_2 + c_3 A_3 + c_4 A_4 = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = A.$$

A is zero only if the c ’s are all zero—this proves independence of A_1, A_2, A_3, A_4 .

The three matrices A_1, A_2, A_4 are a basis for a subspace—the upper triangular matrices. Its dimension is 3. A_1 and A_4 are a basis for the diagonal matrices. What is a basis for the symmetric matrices? Keep A_1 and A_4 , and throw in $A_2 + A_3$.

To push this further, think about the space of all n by n matrices. One possible basis uses matrices that have only a single nonzero entry (that entry is 1). There are n^2 positions for that 1, so there are n^2 basis matrices:

The dimension of the whole n by n matrix space is n^2 .

The dimension of the subspace of *upper triangular* matrices is $\frac{1}{2}n^2 + \frac{1}{2}n$.

The dimension of the subspace of *diagonal* matrices is n .

The dimension of the subspace of *symmetric* matrices is $\frac{1}{2}n^2 + \frac{1}{2}n$ (why?).

Function spaces The equations $d^2y/dx^2 = 0$ and $d^2y/dx^2 = -y$ and $d^2y/dx^2 = y$ involve the second derivative. In calculus we solve to find the functions $y(x)$:

$$\begin{array}{ll} y'' = 0 & \text{is solved by any linear function } y = cx + d \\ y'' = -y & \text{is solved by any combination } y = c \sin x + d \cos x \\ y'' = y & \text{is solved by any combination } y = ce^x + de^{-x}. \end{array}$$

That solution space for $y'' = -y$ has two basis functions: $\sin x$ and $\cos x$. The space for $y'' = 0$ has x and 1. It is the “nullspace” of the second derivative! The dimension is 2 in each case (these are second-order equations).

The solutions of $y'' = 2$ don't form a subspace—the right side $b = 2$ is not zero. A particular solution is $y(x) = x^2$. The complete solution is $y(x) = x^2 + cx + d$. All those functions satisfy $y'' = 2$. Notice the particular solution plus any function $cx + d$ in the nullspace. A linear differential equation is like a linear matrix equation $Ax = b$. But we solve it by calculus instead of linear algebra.

We end here with the space $\mathbf{Z}$ that contains only the zero vector. The dimension of this space is *zero*. **The empty set** (containing no vectors) **is a basis for $\mathbf{Z}$** . We can never allow the zero vector into a basis, because then linear independence is lost.

■ REVIEW OF THE KEY IDEAS ■

1. The columns of A are *independent* if $x = \mathbf{0}$ is the only solution to $Ax = \mathbf{0}$.
2. The vectors $v_1, \dots, v_r$ *span* a space if their combinations fill that space.
3. **A basis consists of linearly independent vectors that span the space.** Every vector in the space is a *unique* combination of the basis vectors.
4. All bases for a space have the same number of vectors. This number of vectors in a basis is the *dimension* of the space.
5. The pivot columns are one basis for the column space. The dimension is r .

■ WORKED EXAMPLES ■

3.5 A Start with the vectors $v_1 = (1, 2, 0)$ and $v_2 = (2, 3, 0)$. (a) Are they linearly independent? (b) Are they a basis for any space? (c) What space V do they span? (d) What is the dimension of V ? (e) Which matrices A have V as their column space? (f) Which matrices have V as their nullspace? (g) Describe all vectors v_3 that complete a basis v_1, v_2, v_3 for $\mathbb{R}^3$.

Solution

- (a) v_1 and v_2 are independent—the only combination to give $\mathbf{0}$ is $0v_1 + 0v_2$.
- (b) Yes, they are a basis for the space they span.
- (c) That space V contains all vectors $(x, y, 0)$. It is the xy plane in $\mathbb{R}^3$.
- (d) The dimension of V is 2 since the basis contains two vectors.
- (e) This V is the column space of any 3 by n matrix A of rank 2, if every column is a combination of v_1 and v_2 . In particular A could just have columns v_1 and v_2 .
- (f) This V is the nullspace of any m by 3 matrix B of rank 1, if every row is a multiple of $(0, 0, 1)$. In particular take $B = [0 \ 0 \ 1]$. Then $Bv_1 = \mathbf{0}$ and $Bv_2 = \mathbf{0}$.
- (g) Any third vector $v_3 = (a, b, c)$ will complete a basis for $\mathbb{R}^3$ provided $c \neq 0$.

3.5 B Start with three independent vectors w_1, w_2, w_3 . Take combinations of those vectors to produce v_1, v_2, v_3 . Write the combinations in matrix form as $V = WM$:

$$\begin{aligned} v_1 &= w_1 + w_2 \\ v_2 &= w_1 + 2w_2 + w_3 \\ v_3 &= w_2 + cw_3 \end{aligned} \quad \text{which is} \quad \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & c \end{bmatrix}$$

What is the test on a matrix V to see if its columns are linearly independent? If $c \neq 1$ show that v_1, v_2, v_3 are linearly independent. If $c = 1$ show that the v 's are linearly *dependent*.

Solution The test on V for independence of its columns was in our first definition: *The nullspace of V must contain only the zero vector.* Then $x = (0, 0, 0)$ is the only combination of the columns that gives $Vx = \text{zero vector}$.

If $c = 1$ in our problem, we can see *dependence* in two ways. First, $v_1 + v_3$ will be the same as v_2 . (If you add $w_1 + w_2$ to $w_2 + w_3$ you get $w_1 + 2w_2 + w_3$ which is v_2 .) In other words $v_1 - v_2 + v_3 = \mathbf{0}$ —which says that the v 's are not independent.

The other way is to look at the nullspace of M . If $c = 1$, the vector $x = (1, -1, 1)$ is in that nullspace, and $Mx = \mathbf{0}$. Then certainly $WMx = \mathbf{0}$ which is the same as $Vx = \mathbf{0}$. So the v 's are dependent. This specific $x = (1, -1, 1)$ from the nullspace tells us again that $v_1 - v_2 + v_3 = \mathbf{0}$.

Now suppose $c \neq 1$. Then the matrix M is invertible. So if x is *any nonzero vector* we know that Mx is nonzero. Since the w 's are given as independent, we further know that WMx is nonzero. Since $V = WM$, this says that x is *not* in the nullspace of V . In other words v_1, v_2, v_3 are independent.

The general rule is “independent v 's from independent w 's when M is invertible”. And if these vectors are in $\mathbf{R}^3$, they are not only independent—they are a basis for $\mathbf{R}^3$. “Basis of v 's from basis of w 's when the change of basis matrix M is invertible.”

3.5 C (Important example) Suppose $v_1, \dots, v_n$ is a basis for $\mathbf{R}^n$ and the n by n matrix A is invertible. Show that $Av_1, \dots, Av_n$ is also a basis for $\mathbf{R}^n$.

Solution In *matrix language*: Put the basis vectors $v_1, \dots, v_n$ in the columns of an invertible(!) matrix V . Then $Av_1, \dots, Av_n$ are the columns of AV . Since A is invertible, so is AV and its columns give a basis.

In *vector language*: Suppose $c_1Av_1 + \dots + c_nAv_n = \mathbf{0}$. This is $Av = \mathbf{0}$ with $v = c_1v_1 + \dots + c_nv_n$. Multiply by A^{-1} to reach $v = \mathbf{0}$. By linear independence of the v 's, all $c_i = 0$. This shows that the Av 's are independent.

To show that the Av 's span $\mathbf{R}^n$, solve $c_1Av_1 + \dots + c_nAv_n = b$ which is the same as $c_1v_1 + \dots + c_nv_n = A^{-1}b$. Since the v 's are a basis, this must be solvable.

Problem Set 3.5

Questions 1–10 are about linear independence and linear dependence.

- 1 Show that v_1, v_2, v_3 are independent but v_1, v_2, v_3, v_4 are dependent:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad v_4 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

Solve $c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = \mathbf{0}$ or $Ax = \mathbf{0}$. The v 's go in the columns of A .

- 2 (Recommended) Find the largest possible number of independent vectors among

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \quad v_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \quad v_5 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \quad v_6 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

- 3 Prove that if $a = 0$ or $d = 0$ or $f = 0$ (3 cases), the columns of U are dependent:

$$U = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}.$$

- 4 If a, d, f in Question 3 are all nonzero, show that the only solution to $Ux = 0$ is $x = 0$. Then the upper triangular U has independent columns.
- 5 Decide the dependence or independence of
- the vectors $(1, 3, 2)$ and $(2, 1, 3)$ and $(3, 2, 1)$
 - the vectors $(1, -3, 2)$ and $(2, 1, -3)$ and $(-3, 2, 1)$.
- 6 Choose three independent columns of U . Then make two other choices. Do the same for A .

$$U = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 6 & 7 & 0 \\ 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 6 & 7 & 0 \\ 0 & 0 & 0 & 9 \\ 4 & 6 & 8 & 2 \end{bmatrix}.$$

- 7 If w_1, w_2, w_3 are independent vectors, show that the differences $v_1 = w_2 - w_3$ and $v_2 = w_1 - w_3$ and $v_3 = w_1 - w_2$ are *dependent*. Find a combination of the v 's that gives zero. Which matrix A in $[v_1 \ v_2 \ v_3] = [w_1 \ w_2 \ w_3] A$ is singular?
- 8 If w_1, w_2, w_3 are independent vectors, show that the sums $v_1 = w_2 + w_3$ and $v_2 = w_1 + w_3$ and $v_3 = w_1 + w_2$ are *independent*. (Write $c_1v_1 + c_2v_2 + c_3v_3 = 0$ in terms of the w 's. Find and solve equations for the c 's, to show they are zero.)
- 9 Suppose v_1, v_2, v_3, v_4 are vectors in $\mathbf{R}^3$.
- These four vectors are dependent because _____.
 - The two vectors v_1 and v_2 will be dependent if _____.
 - The vectors v_1 and $(0, 0, 0)$ are dependent because _____.
- 10 Find two independent vectors on the plane $x + 2y - 3z - t = 0$ in $\mathbf{R}^4$. Then find three independent vectors. Why not four? This plane is the nullspace of what matrix?

Questions 11–15 are about the space *spanned* by a set of vectors. Take all linear combinations of the vectors.

- 11 Describe the subspace of $\mathbf{R}^3$ (is it a line or plane or $\mathbf{R}^3$?) spanned by
- the two vectors $(1, 1, -1)$ and $(-1, -1, 1)$
 - the three vectors $(0, 1, 1)$ and $(1, 1, 0)$ and $(0, 0, 0)$
 - all vectors in $\mathbf{R}^3$ with whole number components
 - all vectors with positive components.
- 12 The vector b is in the subspace spanned by the columns of A when _____ has a solution. The vector c is in the row space of A when _____ has a solution.
- True or false:* If the zero vector is in the row space, the rows are dependent.

- 13 Find the dimensions of these 4 spaces. Which two of the spaces are the same? (a) column space of A , (b) column space of U , (c) row space of A , (d) row space of U :

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

- 14 $v + w$ and $v - w$ are combinations of v and w . Write v and w as combinations of $v + w$ and $v - w$. The two pairs of vectors _____ the same space. When are they a basis for the same space?

Questions 15–25 are about the requirements for a basis.

- 15 If $v_1, \dots, v_n$ are linearly independent, the space they span has dimension _____. These vectors are a _____ for that space. If the vectors are the columns of an m by n matrix, then m is _____ than n . If $m = n$, that matrix is _____.
- 16 Find a basis for each of these subspaces of $\mathbf{R}^4$:
- (a) All vectors whose components are equal.
 - (b) All vectors whose components add to zero.
 - (c) All vectors that are perpendicular to $(1, 1, 0, 0)$ and $(1, 0, 1, 1)$.
 - (d) The column space and the nullspace of I (4 by 4).
- 17 Find three different bases for the column space of $U = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$. Then find two different bases for the row space of U .
- 18 Suppose $v_1, v_2, \dots, v_6$ are six vectors in $\mathbf{R}^4$.
- (a) Those vectors (do)(do not)(might not) span $\mathbf{R}^4$.
 - (b) Those vectors (are)(are not)(might be) linearly independent.
 - (c) Any four of those vectors (are)(are not)(might be) a basis for $\mathbf{R}^4$.
- 19 The columns of A are n vectors from $\mathbf{R}^m$. If they are linearly independent, what is the rank of A ? If they span $\mathbf{R}^m$, what is the rank? If they are a basis for $\mathbf{R}^m$, what then? *Looking ahead:* The rank r counts the number of _____ columns.
- 20 Find a basis for the plane $x - 2y + 3z = 0$ in $\mathbf{R}^3$. Then find a basis for the intersection of that plane with the xy plane. Then find a basis for all vectors perpendicular to the plane.
- 21 Suppose the columns of a 5 by 5 matrix A are a basis for $\mathbf{R}^5$.
- (a) The equation $Ax = \mathbf{0}$ has only the solution $x = \mathbf{0}$ because _____.
 - (b) If b is in $\mathbf{R}^5$ then $Ax = b$ is solvable because the basis vectors _____ $\mathbf{R}^5$.

Conclusion: A is invertible. Its rank is 5. Its rows are also a basis for $\mathbf{R}^5$.

- 22** Suppose S is a 5-dimensional subspace of $\mathbf{R}^6$. True or false (example if false):
- (a) Every basis for S can be extended to a basis for $\mathbf{R}^6$ by adding one more vector.
 - (b) Every basis for $\mathbf{R}^6$ can be reduced to a basis for S by removing one vector.
- 23** U comes from A by subtracting row 1 from row 3:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Find bases for the two column spaces. Find bases for the two row spaces. Find bases for the two nullspaces. Which spaces stay fixed in elimination?

- 24** True or false (give a good reason):
- (a) If the columns of a matrix are dependent, so are the rows.
 - (b) The column space of a 2 by 2 matrix is the same as its row space.
 - (c) The column space of a 2 by 2 matrix has the same dimension as its row space.
 - (d) The columns of a matrix are a basis for the column space.
- 25** For which numbers c and d do these matrices have rank 2?

$$A = \begin{bmatrix} 1 & 2 & 5 & 0 & 5 \\ 0 & 0 & c & 2 & 2 \\ 0 & 0 & 0 & d & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} c & d \\ d & c \end{bmatrix}.$$

Questions 26–30 are about spaces where the “vectors” are matrices.

- 26** Find a basis (and the dimension) for each of these subspaces of 3 by 3 matrices:
- (a) All diagonal matrices.
 - (b) All symmetric matrices ($A^T = A$).
 - (c) All skew-symmetric matrices ($A^T = -A$).
- 27** Construct six linearly independent 3 by 3 echelon matrices $U_1, \dots, U_6$.
- 28** Find a basis for the space of all 2 by 3 matrices whose columns add to zero. Find a basis for the subspace whose rows also add to zero.
- 29** What subspace of 3 by 3 matrices is spanned (take all combinations) by
- (a) the invertible matrices?
 - (b) the rank one matrices?
 - (c) the identity matrix?
- 30** Find a basis for the space of 2 by 3 matrices whose nullspace contains $(2, 1, 1)$.

Questions 31–35 are about spaces where the “vectors” are functions.

- 31 (a) Find all functions that satisfy $\frac{dy}{dx} = 0$.
 (b) Choose a particular function that satisfies $\frac{dy}{dx} = 3$.
 (c) Find all functions that satisfy $\frac{dy}{dx} = 3$.
- 32 The cosine space $\mathbf{F}_3$ contains all combinations $y(x) = A \cos x + B \cos 2x + C \cos 3x$. Find a basis for the subspace with $y(0) = 0$.
- 33 Find a basis for the space of functions that satisfy
 (a) $\frac{dy}{dx} - 2y = 0$
 (b) $\frac{dy}{dx} - \frac{y}{x} = 0$.
- 34 Suppose $y_1(x)$, $y_2(x)$, $y_3(x)$ are three different functions of x . The vector space they span could have dimension 1, 2, or 3. Give an example of y_1, y_2, y_3 to show each possibility.
- 35 Find a basis for the space of polynomials $p(x)$ of degree ≤ 3 . Find a basis for the subspace with $p(1) = 0$.
- 36 Find a basis for the space $\mathbf{S}$ of vectors (a, b, c, d) with $a + c + d = 0$ and also for the space $\mathbf{T}$ with $a + b = 0$ and $c = 2d$. What is the dimension of the intersection $\mathbf{S} \cap \mathbf{T}$?
- 37 If $AS = SA$ for the *shift matrix* S , show that A must have this special form:

$$\text{If } \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \text{ then } A = \begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}.$$

“The subspace of matrices that commute with the shift S has dimension ____.”

- 38 Which of the following are bases for $\mathbf{R}^3$?
 (a) $(1, 2, 0)$ and $(0, 1, -1)$
 (b) $(1, 1, -1)$, $(2, 3, 4)$, $(4, 1, -1)$, $(0, 1, -1)$
 (c) $(1, 2, 2)$, $(-1, 2, 1)$, $(0, 8, 0)$
 (d) $(1, 2, 2)$, $(-1, 2, 1)$, $(0, 8, 6)$
- 39 Suppose A is 5 by 4 with rank 4. Show that $A\mathbf{x} = \mathbf{b}$ has no solution when the 5 by 5 matrix $[A \ \mathbf{b}]$ is invertible. Show that $A\mathbf{x} = \mathbf{b}$ is solvable when $[A \ \mathbf{b}]$ is singular.
- 40 (a) Find a basis for all solutions to $d^4y/dx^4 = y(x)$.
 (b) Find a particular solution to $d^4y/dx^4 = y(x) + 1$. Find the complete solution.

Challenge Problems

- 41** Write the 3 by 3 identity matrix as a combination of the other five permutation matrices! Then show that those five matrices are linearly independent. (Assume a combination gives $c_1 P_1 + \cdots + c_5 P_5 =$ zero matrix, and check entries to prove c_i is zero.) The five permutations are a basis for the subspace of 3 by 3 matrices with row and column sums all equal.
- 42** Choose $\mathbf{x} = (x_1, x_2, x_3, x_4)$ in $\mathbf{R}^4$. It has 24 rearrangements like (x_2, x_1, x_3, x_4) and (x_4, x_3, x_1, x_2) . Those 24 vectors, including $\mathbf{x}$ itself, span a subspace $\mathbf{S}$. Find specific vectors $\mathbf{x}$ so that the dimension of $\mathbf{S}$ is: (a) zero, (b) one, (c) three, (d) four.
- 43** Intersections and sums have $\dim(\mathbf{V}) + \dim(\mathbf{W}) = \dim(\mathbf{V} \cap \mathbf{W}) + \dim(\mathbf{V} + \mathbf{W})$. Start with a basis $\mathbf{u}_1, \dots, \mathbf{u}_r$ for the intersection $\mathbf{V} \cap \mathbf{W}$. Extend with $\mathbf{v}_1, \dots, \mathbf{v}_s$ to a basis for $\mathbf{V}$, and separately with $\mathbf{w}_1, \dots, \mathbf{w}_t$ to a basis for $\mathbf{W}$. Prove that the $\mathbf{u}$'s, $\mathbf{v}$'s and $\mathbf{w}$'s together are *independent*. The dimensions have $(r + s) + (r + t) = (r) + (r + s + t)$ as desired.
- 44** Mike Artin suggested a neat higher-level proof of that dimension formula in Problem 43. From all inputs $\mathbf{v}$ in $\mathbf{V}$ and $\mathbf{w}$ in $\mathbf{W}$, the “sum transformation” produces $\mathbf{v} + \mathbf{w}$. Those outputs fill the space $\mathbf{V} + \mathbf{W}$. The nullspace contains all pairs $\mathbf{v} = \mathbf{u}$, $\mathbf{w} = -\mathbf{u}$ for vectors $\mathbf{u}$ in $\mathbf{V} \cap \mathbf{W}$. (Then $\mathbf{v} + \mathbf{w} = \mathbf{u} - \mathbf{u} = \mathbf{0}$.) So $\dim(\mathbf{V} + \mathbf{W}) + \dim(\mathbf{V} \cap \mathbf{W})$ equals $\dim(\mathbf{V}) + \dim(\mathbf{W})$ (*input dimension from $\mathbf{V}$ and $\mathbf{W}$*) by the crucial formula

$$\text{dimension of outputs} + \text{dimension of nullspace} = \text{dimension of inputs}.$$

Problem For an m by n matrix of rank r , what are those 3 dimensions? Outputs = column space. This question will be answered in Section 3.6, can you do it now?

- 45** Inside $\mathbf{R}^n$, suppose $\dim(\mathbf{V}) + \dim(\mathbf{W}) > n$. Show that some nonzero vector is in both $\mathbf{V}$ and $\mathbf{W}$.
- 46** Suppose A is 10 by 10 and $A^2 = 0$ (zero matrix). This means that the column space of A is contained in the _____. If A has rank r , those subspaces have dimension $r \leq 10 - r$. So the rank is $r \leq 5$.

(This problem was added to the second printing: If $A^2 = 0$ it says that $r \leq n/2$.)

3.6 Dimensions of the Four Subspaces

The main theorem in this chapter connects **rank** and **dimension**. The **rank** of a matrix is the number of pivots. The **dimension** of a subspace is the number of vectors in a basis. We count pivots or we count basis vectors. *The rank of A reveals the dimensions of all four fundamental subspaces.* Here are the subspaces, including the new one.

Two subspaces come directly from A , and the other two from A^T :

Four Fundamental Subspaces

1. The **row space** is $C(A^T)$, a subspace of $\mathbf{R}^n$.
2. The **column space** is $C(A)$, a subspace of $\mathbf{R}^m$.
3. The **nullspace** is $N(A)$, a subspace of $\mathbf{R}^n$.
4. The **left nullspace** is $N(A^T)$, a subspace of $\mathbf{R}^m$. This is our new space.

In this book the column space and nullspace came first. We know $C(A)$ and $N(A)$ pretty well. Now the other two subspaces come forward. The row space contains all combinations of the rows. *This is the column space of A^T .*

For the left nullspace we solve $A^T y = 0$ —that system is n by m . *This is the nullspace of A^T .* The vectors y go on the *left* side of A when the equation is written as $y^T A = 0^T$. The matrices A and A^T are usually different. So are their column spaces and their nullspaces. But those spaces are connected in an absolutely beautiful way.

Part 1 of the Fundamental Theorem finds the dimensions of the four subspaces. One fact stands out: **The row space and column space have the same dimension r** (the rank of the matrix). The other important fact involves the two nullspaces:

$N(A)$ and $N(A^T)$ have dimensions $n - r$ and $m - r$, to make up the full n and m .

Part 2 of the Fundamental Theorem will describe how the four subspaces fit together (two in $\mathbf{R}^n$ and two in $\mathbf{R}^m$). That completes the “right way” to understand every $Ax = b$. Stay with it—you are doing real mathematics.

The Four Subspaces for R

Suppose A is reduced to its row echelon form R . For that special form, the four subspaces are easy to identify. We will find a basis for each subspace and check its dimension. Then we watch how the subspaces change (two of them don’t change!) as we look back at A . The main point is that *the four dimensions are the same for A and R .*

As a specific 3 by 5 example, look at the four subspaces for the echelon matrix R :

$$\begin{array}{lll} m = 3 & \begin{bmatrix} 1 & 3 & 5 & 0 & 7 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \text{pivot rows 1 and 2} \\ n = 5 & & \\ r = 2 & & \text{pivot columns 1 and 4} \end{array}$$

The rank of this matrix R is $r = 2$ (two pivots). Take the four subspaces in order.

1. The **row space** of R has dimension 2, matching the rank.

Reason: The first two rows are a basis. The row space contains combinations of all three rows, but the third row (the zero row) adds nothing new. So rows 1 and 2 span the row space $C(R^T)$.

The pivot rows 1 and 2 are independent. That is obvious for this example, and it is always true. If we look only at the pivot columns, we see the r by r identity matrix. There is no way to combine its rows to give the zero row (except by the combination with all coefficients zero). So the r pivot rows are a basis for the row space.

The dimension of the row space is the rank r . The nonzero rows of R form a basis.

2. The **column space** of R also has dimension $r = 2$.

Reason: The pivot columns 1 and 4 form a basis for $C(R)$. They are independent because they start with the r by r identity matrix. No combination of those pivot columns can give the zero column (except the combination with all coefficients zero). And they also span the column space. Every other (free) column is a combination of the pivot columns. Actually the combinations we need are the three special solutions!

Column 2 is 3 (column 1). The special solution is $(-3, 1, 0, 0, 0)$.

Column 3 is 5 (column 1). The special solution is $(-5, 0, 1, 0, 0)$.

Column 5 is 7 (column 1) + 2 (column 4). That solution is $(-7, 0, 0, -2, 1)$.

The pivot columns are independent, and they span, so they are a basis for $C(R)$.

The dimension of the column space is the rank r . The pivot columns form a basis.

3. The **nullspace** has dimension $n - r = 5 - 2$. There are $n - r = 3$ free variables. Here x_2, x_3, x_5 are free (no pivots in those columns). They yield the three special solutions to $Rx = 0$. Set a free variable to 1, and solve for x_1 and x_4 :

$$s_2 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad s_3 = \begin{bmatrix} -5 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad s_5 = \begin{bmatrix} -7 \\ 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} \quad \begin{array}{l} Rx = 0 \text{ has the} \\ \text{complete solution} \\ x = x_2 s_2 + x_3 s_3 + x_5 s_5 \end{array}$$

There is a special solution for each free variable. With n variables and r pivot variables, that leaves $n - r$ free variables and special solutions. $N(R)$ has dimension $n - r$.

The nullspace has dimension $n - r$. The special solutions form a basis.

The special solutions are independent, because they contain the identity matrix in rows 2, 3, 5. All solutions are combinations of special solutions, $x = x_2s_2 + x_3s_3 + x_5s_5$, because this puts x_2 , x_3 and x_5 in the correct positions. Then the pivot variables x_1 and x_4 are totally determined by the equations $Rx = 0$.

4. The nullspace of R^T (left nullspace of R) has dimension $m - r = 3 - 2$.

Reason: The equation $R^T y = 0$ looks for combinations of the columns of R^T (the rows of R) that produce zero. This equation $R^T y = 0$ or $y^T R = 0^T$ is

$$\begin{array}{rcl} \text{Left nullspace} & \begin{array}{l} y_1 [1, 3, 5, 0, 7] \\ + y_2 [0, 0, 0, 1, 2] \\ + y_3 [0, 0, 0, 0, 0] \\ \hline [0, 0, 0, 0, 0] \end{array} & (1) \end{array}$$

The solutions y_1, y_2, y_3 are pretty clear. We need $y_1 = 0$ and $y_2 = 0$. The variable y_3 is free (it can be anything). The nullspace of R^T contains all vectors $y = (0, 0, y_3)$. It is the line of all multiples of the basis vector $(0, 0, 1)$.

In all cases R ends with $m - r$ zero rows. Every combination of these $m - r$ rows gives zero. These are the *only* combinations of the rows of R that give zero, because the pivot rows are linearly independent. The left nullspace of R contains all these solutions $y = (0, \dots, 0, y_{r+1}, \dots, y_m)$ to $R^T y = 0$.

If A is m by n of rank r , its left nullspace has dimension $m - r$.

To produce a zero combination, y must start with r zeros. This leaves dimension $m - r$.

Why is this a “left nullspace”? The reason is that $R^T y = 0$ can be transposed to $y^T R = 0^T$. Now y^T is a row vector to the *left* of R . You see the y ’s in equation (1) multiplying the rows. This subspace came fourth, and some linear algebra books omit it—but that misses the beauty of the whole subject.

In R^n the row space and nullspace have dimensions r and $n - r$ (adding to n).

In R^m the column space and left nullspace have dimensions r and $m - r$ (total m).

So far this is proved for echelon matrices R . Figure 3.5 shows the same for A .

The Four Subspaces for A

We have a job still to do. *The subspace dimensions for A are the same as for R .* The job is to explain why. A is now any matrix that reduces to $R = \text{rref}(A)$.

$$A \text{ reduces to } R \quad A = \begin{bmatrix} 1 & 3 & 5 & 0 & 7 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & 3 & 5 & 1 & 9 \end{bmatrix} \quad \text{Notice } C(A) \neq C(R) \quad (2)$$

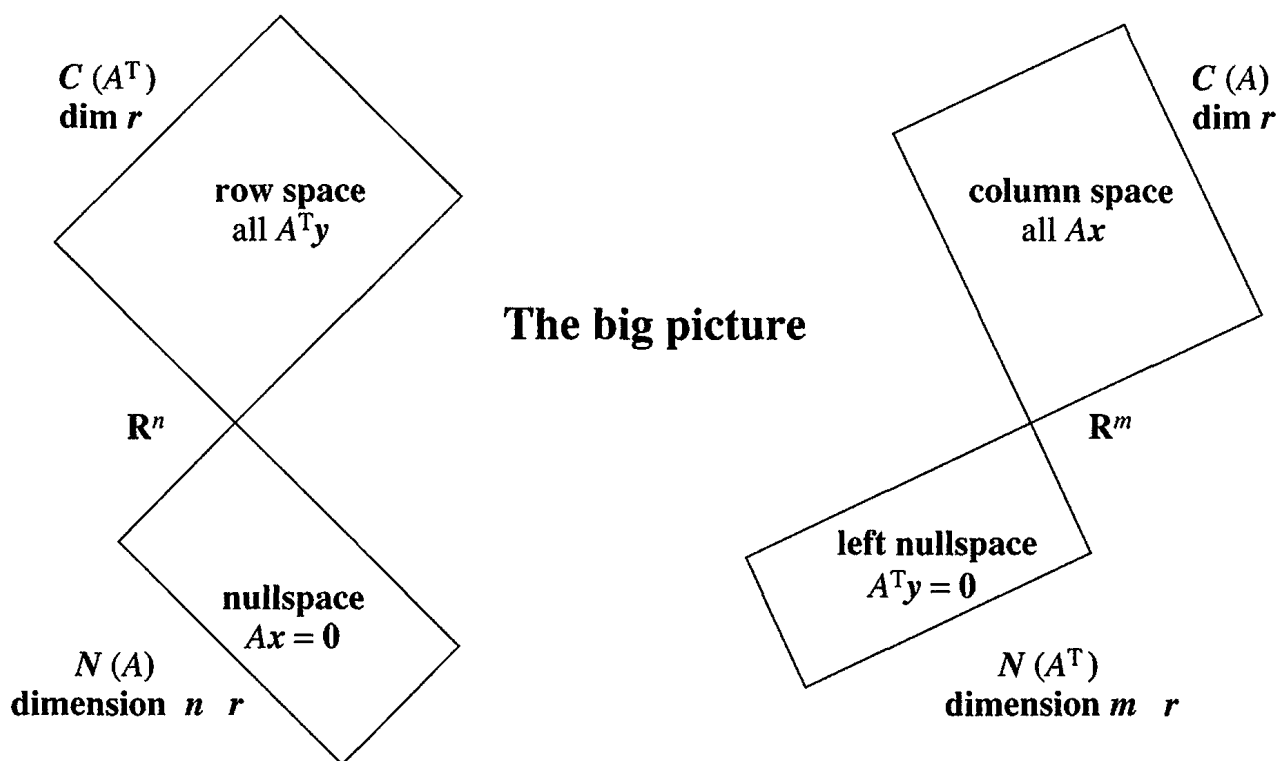


Figure 3.5: The dimensions of the Four Fundamental Subspaces (for R and for A).

An elimination matrix takes A to R . The big picture (Figure 3.5) applies to both. The invertible matrix E is the product of the elementary matrices that reduce A to R :

$$A \text{ to } R \text{ and back} \quad EA = R \quad \text{and} \quad A = E^{-1}R \quad (3)$$

1 A has the same row space as R . Same dimension r and same basis.

Reason: Every row of A is a combination of the rows of R . Also every row of R is a combination of the rows of A . Elimination changes rows, but not row spaces.

Since A has the same row space as R , we can choose the first r rows of R as a basis. Or we could choose r suitable rows of the original A . They might not always be the *first* r rows of A , because those could be dependent. The good r rows of A are the ones that end up as pivot rows in R .

2 The column space of A has dimension r . For every matrix this is essential:

The number of independent columns equals the number of independent rows.

Wrong reason: “ A and R have the same column space.” This is false. The columns of R often end in zeros. The columns of A don’t often end in zeros. The column spaces are different, but their *dimensions* are the same—equal to r .

Right reason: The *same* combinations of the columns are zero (or nonzero) for A and R . Say that another way: $Ax = 0$ exactly when $Rx = 0$. The r pivot columns (of both) are independent.

Conclusion The r pivot columns of A are a basis for its column space.

3 *A has the same nullspace as R. Same dimension $n - r$ and same basis.*

Reason: The elimination steps don't change the solutions. The special solutions are a basis for this nullspace (as we always knew). There are $n - r$ free variables, so the dimension of the nullspace is $n - r$. Notice that $r + (n - r)$ equals n :

$$(\text{dimension of column space}) + (\text{dimension of nullspace}) = \text{dimension of } \mathbf{R}^n.$$

4 *The left nullspace of A (the nullspace of A^T) has dimension $m - r$.*

Reason: A^T is just as good a matrix as A . When we know the dimensions for every A , we also know them for A^T . Its column space was proved to have dimension r . Since A^T is n by m , the "whole space" is now $\mathbf{R}^m$. The counting rule for A was $r + (n - r) = n$. The counting rule for A^T is $r + (m - r) = m$. We now have all details of the main theorem:

Fundamental Theorem of Linear Algebra, Part 1

The column space and row space both have dimension r .

The nullspaces have dimensions $n - r$ and $m - r$.

By concentrating on *spaces* of vectors, not on individual numbers or vectors, we get these clean rules. You will soon take them for granted—eventually they begin to look obvious. But if you write down an 11 by 17 matrix with 187 nonzero entries, I don't think most people would see why these facts are true:

Two key facts $\text{dimension of } C(A) = \text{dimension of } C(A^T) = \text{rank of } A$
 $\text{dimension of } C(A) + \text{dimension of } N(A) = 17.$

Example 1 $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ has $m = 1$ and $n = 3$ and rank $r = 1$.

The row space is a line in $\mathbf{R}^3$. The nullspace is the plane $Ax = x_1 + 2x_2 + 3x_3 = 0$. This plane has dimension 2 (which is $3 - 1$). The dimensions add to $1 + 2 = 3$.

The columns of this 1 by 3 matrix are in $\mathbf{R}^1$! The column space is all of $\mathbf{R}^1$. The left nullspace contains only the zero vector. The only solution to $A^T y = 0$ is $y = 0$, no other multiple of $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ gives the zero row. Thus $N(A^T)$ is $\mathbf{Z}$, the zero space with dimension 0 (which is $m - r$). In $\mathbf{R}^m$ the dimensions add to $1 + 0 = 1$.

Example 2 $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$ has $m = 2$ with $n = 3$ and rank $r = 1$.

The row space is the same line through $(1, 2, 3)$. The nullspace must be the same plane $x_1 + 2x_2 + 3x_3 = 0$. Their dimensions still add to $1 + 2 = 3$.

All columns are multiples of the first column $(1, 2)$. Twice the first row minus the second row is the zero row. Therefore $A^T y = 0$ has the solution $y = (2, -1)$. The column space and left nullspace are **perpendicular lines** in $\mathbf{R}^2$. Dimensions $1 + 1 = 2$.

Column space = line through $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ Left nullspace = line through $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

If A has three equal rows, its rank is _____. What are two of the y 's in its left nullspace?

The y 's in the left nullspace combine the rows to give the zero row.

Matrices of Rank One

That last example had rank $r = 1$ —and rank one matrices are special. We can describe them all. You will see again that dimension of row space = dimension of column space. When $r = 1$, every row is a multiple of the same row:

$$A = uv^T \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ -3 & -6 & -9 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{equals} \quad \begin{bmatrix} 1 \\ 2 \\ -3 \\ 0 \end{bmatrix} \quad \text{times} \quad [1 \ 2 \ 3] = v^T.$$

A column times a row (4 by 1 times 1 by 3) produces a matrix (4 by 3). All rows are multiples of the row (1, 2, 3). All columns are multiples of the column (1, 2, −3, 0). The row space is a line in $\mathbf{R}^n$, and the column space is a line in $\mathbf{R}^m$.

Every rank one matrix has the special form $A = uv^T$ = column times row.

The columns are multiples of u . The rows are multiples of v^T . *The nullspace is the plane perpendicular to v .* ($Ax = 0$ means that $u(v^T x) = 0$ and then $v^T x = 0$.) It is this **perpendicularity** of the subspaces that will be Part 2 of the Fundamental Theorem.

■ REVIEW OF THE KEY IDEAS ■

1. The r pivot rows of R are a basis for the row spaces of R and A (same space).
2. The r pivot columns of A (!) are a basis for its column space.
3. The $n - r$ special solutions are a basis for the nullspaces of A and R (same space).
4. The last $m - r$ rows of I are a basis for the left nullspace of R .
5. The last $m - r$ rows of E are a basis for the left nullspace of A .

Note about the four subspaces The Fundamental Theorem looks like pure algebra, but it has very important applications. My favorites are the networks in Chapter 8 (often I go there for my next lecture). The equation for y in the left nullspace is $A^T y = 0$:

Flow into a node equals flow out. Kirchhoff's Current Law is the "balance equation".

This is (in my opinion) the most important equation in applied mathematics. All models in science and engineering and economics involve a balance—of force or heat flow or charge or momentum or money. That balance equation, plus Hooke's Law or Ohm's Law or some law connecting "potentials" to "flows", gives a clear framework for applied mathematics.

My textbook on *Computational Science and Engineering* develops that framework, together with algorithms to solve the equations: Finite differences, finite elements, spectral methods, iterative methods, and multigrid.